# Expressive Mapper Pro 3

Official User Manual & Algorithmic Specification — Synthesizer V Studio 2 PRO

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**Version:** v3.6.2 (DOD Engine)

**Minimum Environment:** SynthV Studio 2 PRO v2.2.1+ (Build 67072)

**Memory Allocation:** Low Dynamic Allocation (Low GC)

This manual presents comprehensive technical documentation for **Expressive Mapper Pro 3**. Each module is organized into dedicated sections for each standalone `.lua` script, as well as two levels of depth: a **Simple Summary** for fast workflow understanding and an **Extended Technical Specification** for advanced producers and audio engineers.

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## 1. INSTALLATION & SYSTEM REQUIREMENTS

### SIMPLE SUMMARY (PRACTICAL USAGE)

To install the suite or any individual `.lua` script in Synthesizer V Studio Pro:

- Open Synthesizer V Studio and navigate to top menu `Scripts` → `Open Scripts Folder`.
- Copy the desired compiled `.lua` files (`Expressive_Lyric_Melody.lua`, `Expressive_Vocal_Automation.lua`, `Expressive_Harmonies.lua`, `Expressive_Chords.lua`) into that directory.
- In SynthV, select `Scripts` → `Rescan Scripts`. The side panel scripts will be available immediately.

### EXTENDED TECHNICAL SPECIFICATION

Execution environment, lifecycle and Dreamtonics API integration:

- **Engine Architecture:** Developed on Lua 5.4 / LuaJIT in `SVClient` client layer as standalone `SidePanelSection` modules. Each script includes an isolated UI state machine.
- **Configuration Persistence:** Reads and writes user interface preferences into a structured JSON file at `%APPDATA%\\Dreamtonics\\Synthesizer V Studio 2\\scripts\\mapeador_user_config.json`.

**Structure of the JSON Configuration File:**

| JSON Parameter | Type | Default Value | Purpose / UI Control |
|---|---|---|---|
| `"idiomaUI"` | Numeric (Int) | `0` (Spanish) | UI language translation (0: ES, 1: EN, 2: JA). |
| `"modo"` | Numeric (Int) | `0` | Active operation mode. |
| `"preset"` | Numeric (Int) | `0` | Index of selected vocal expression preset. |
| `"intensidad"` | Numeric (Int) | `100` | Percentage multiplier for automations (0% to 200%). |

```
[Synthesizer V Scripts Directory]
 └── %APPDATA%\\Dreamtonics\\Synthesizer V Studio 2\\
      └── scripts\\
           ├── Expressive_Lyric_Melody.lua        <-- Panel 1: Prosodic Lyric & Melody
           ├── Expressive_Vocal_Automation.lua    <-- Panel 2: Hermite/TCB Curves
           ├── Expressive_Harmonies.lua           <-- Panel 3: Vocal Harmonies & Just Intonation
           ├── Expressive_Chords.lua              <-- Panel 4: Chord Progressions & Counterpoint
           └── mapeador_user_config.json          <-- User configuration persistence JSON file
```

## 2. SCRIPT 1: EXPRESSIVE_LYRIC_MELODY.LUA (I)

### SIMPLE SUMMARY (WORKFLOW)

This side panel allows you to enter song lyrics and generate vocal melodies immediately. Write your text in the box, choose a musical scale, root key, and melodic contour, then click `Apply` to generate natural-sounding notes.

### TECHNICAL OPERATION SPECIFICATION

Supports two distinct target modes:

- **Create Mode:** Generates new notes starting from the current playhead position. Note durations are calculated dynamically based on chosen preset.
- **Replace Mode:** Set `targetNotesMode` to 1 (Selected Notes). It respects hand-drawn note timings and durations while replacing lyrics and quantizing notes to the active scale.

## 3. SCRIPT 1: EXPRESSIVE_LYRIC_MELODY.LUA (II)

### SIMPLE SUMMARY (TIPS)

Manage note group timestamps easily:

- Use `%` followed by seconds (e.g. `hello %5 world`) to place phrases at absolute seconds.
- Enable **Chops Mode** to cut notes at 50% duration with clean volume gates for rapid, staccato vocal cuts (Artcore/Breakcore style).

### PROSODIC AND ALGORITHMIC SPECIFICATION

Syllable mapping and non-overlapping note group alignment:

- **Timestamp Markers (%):** Parser extracts markers and creates independent `NoteGroupReference` items. Offset is calculated in absolute blicks, applying `setTimeOffset(startBlick)` and `setTimeRange(startBlick, dur)`.
- **RAE Accents:** Syllables containing Spanish accents (á, é, í, ó, ú) trigger a +20 cents pitch boost and +15% tension to emulate real vocal stress.

```
[Group Segmentation by Timestamp %]
 Input:  "chorus %10 start %24 verse"
 Track:  [Group 1: 0s - 3s] ----> [Group 2: 10s - 14s] ----> [Group 3: 24s - 29s]
 offsets: (0 blicks)              (120,000 blicks at 120BPM) (288,000 blicks at 120BPM)
```

## 4. SCRIPT 2: EXPRESSIVE_VOCAL_AUTOMATION.LUA (I)

**SIMPLE SUMMARY (VOCAL ENHANCER)**

This panel acts as a "vocal makeup" tool. Select existing notes, pick an expression preset (like Belting, Sad, Whisper) and click Apply. The script automatically draws smooth Hermite curves for tension, breathiness, gender, and loudness.

**HERMITE ENGINE TECHNICAL SPECIFICATION**

Geometric calculation of automation nodes:

- **C1 Continuous Curves:** Implements Hermite cubic splines with TCB controls to prevent overshoots. Distributes 5 key control points per note.
- **Modulated Envelopes:** Writes directly to `tension`, `breathiness`, `gender`, `loudness`, and `voicing` curves.

## 5. SCRIPT 2: EXPRESSIVE_VOCAL_AUTOMATION.LUA (II)

**SIMPLE SUMMARY (TIPS)**

The engine detects phonemes automatically to place expressive peaks:

- Use "Clean Previous" to overwrite existing automation curves.
- Enable "Merge Curves" (Merge Mode) to blend new automations with hand-drawn ones.

**PHONEME ALIGNMENT & RDP SPECIFICATION**

Acoustic sync and data density reduction:

- **Vowel Onset Alignment:** Queries `SV:getPhonemesForGroup`. Shifts breathiness and tension peaks exactly to the `vowelOnsetBlick`.
- **RDP Node Reduction:** Uses the Ramer-Douglas-Peucker algorithm to filter redundant envelope points, keeping CPU usage minimal.

## 6. SCRIPT 3: EXPRESSIVE_HARMONIES.LUA (I)

### SIMPLE SUMMARY (VOCAL CHOIRS)

Generates harmony backing tracks (SATB choir, pop trio, duet) automatically based on the scale of the lead track. It clones settings, creates the notes, and lines up the groups in the timeline.

### DIATONIC HARMONIZER TECHNICAL SPECIFICATION

Track routing and note transposition:

- **Interval Parsing:** Parses diatonic ( `d` prefix) and chromatic ( `c` prefix) intervals (e.g. `+3, -5, c+7` ) and renders notes in new tracks.
- **Group Alignment:** Inherits `timeOffset` and `timeRange` from lead tracks, writing them via:
  `mainRef:setTimeOffset(sourceOffset)\nmainRef:setTimeRange(sourceOnset, sourceDur)`

## 7. SCRIPT 3: EXPRESSIVE_HARMONIES.LUA (II)

**SIMPLE SUMMARY (TUNING & IA)**

To keep backing vocals clean:

- **Anti-phase Detuning:** Applies micro-shifts in pitch and timing to prevent phasing.
- **Just Intonation:** Micro-tunes intervals (Maj/Min 3rds) for pure harmonic resonance.

**PURE TUNING AND AI RETAKES SPECIFICATION**

Microtonal adjustments and AI variations:

- **Just Intonation:** Shifts notes to acoustic ratios (e.g. Major 3rd at -13.69 cents, Minor 3rd at +15.64 cents).
- **AI Retakes:** Automatically triggers `Note.getRetakes():generateTake()` on generated notes to produce unique pronunciations.

## 8. SCRIPT 4: EXPRESSIVE_CHORDS.LUA (I)

**SIMPLE SUMMARY (AUTO-CHORDS)**

Place playhead where you want the music to start, choose a style (Pop, J-Pop Royal, Jazz) and rhythm, and click Apply. The script writes basslines and chord pads using voice leading so that notes move smoothly.

**HARMONIC VOICE LEADING SPECIFICATION**

Minimal energy cost chord inversion:

- **Voice Leading:** Evaluates all candidate chord inversions to minimize total pitch displacement (energy: Sum of [P_new - P_prev]^2).
- **Playhead Sync:** Aligns generated chords to playhead position via `setTimeOffset(startBlick)` and `setTimeRange(startBlick, totalDur)`.

## 9. SCRIPT 4: EXPRESSIVE_CHORDS.LUA (II)

### SIMPLE SUMMARY (COUNTERPOINT & RHYTHMS)

You can generate independent Fuxian counter-melodies that move in opposite direction to the singer. Choose your chord rhythms: long pads, arpeggios, or chops with micro-swing.

### FUXIAN COUNTERPOINT & RHYTHMS SPECIFICATION

Rules and rhythmic swing adjustments:

- **Fux Species:** Evaluates 1st, 2nd, and 3rd species contrapuntal movement. Restricts voice crossing, parallel 5ths, and parallel 8ths.
- **Micro-swing:** Adds subtle time offsets (5% to 20% swing) on weak beats to simulate real keyboard performance.

```
[Voice Leading Optimization]
 Cmaj7 → G4, B4, E5 (Inversion 1)
 Am7   → G4, C5, E5 (Inversion 2)
 Pitch displacement: G4 to G4 (0), B4 to C5 (+1), E5 to E5 (0). Total Cost = 1.
```

## 10. MODE 0: RAE PROSODY & AUTOMATIC EXPRESSION

**SIMPLE SUMMARY (PRACTICAL USAGE)**

Analyzes lyrics to modulate pitch and tension based on speech stress rules. Generates natural fluctuations on stressed syllables and phrase endings.

**EXTENDED TECHNICAL SPECIFICATION**

Prosodic engine and key detection algorithms:

- **Syllabification:** Inspects UTF-8 syllables in Spanish, English, and Japanese. Identifies hiatuses and stressed syllables.
- **Key Detection (Krumhansl-Kessler):** Accumulates a duration-weighted pitch histogram and correlates it with ideal key profiles to determine scale tonic.

**Pearson Correlation Formula for Tonality Detection:**

```
r = Sum((H(i) - H_prom) * (P(i) - P_prom)) / Sqrt[ Sum((H(i) - H_prom)^2 * Sum((P(i) - P_prom)^2) ]
```

Where H(i) represents note durations and P(i) represents ideal key profiles.

## 11. MODE 1: HERMITE CURVES / TCB SPLINES AUTOMATION

### SIMPLE SUMMARY (PRACTICAL USAGE)

Draws smooth parameter curves for tension, breath, and volume instead of sharp steps. Reduces nodes in editor to keep projects clean.

### EXTENDED TECHNICAL SPECIFICATION

TCB Splines and Douglas-Peucker envelope simplifications:

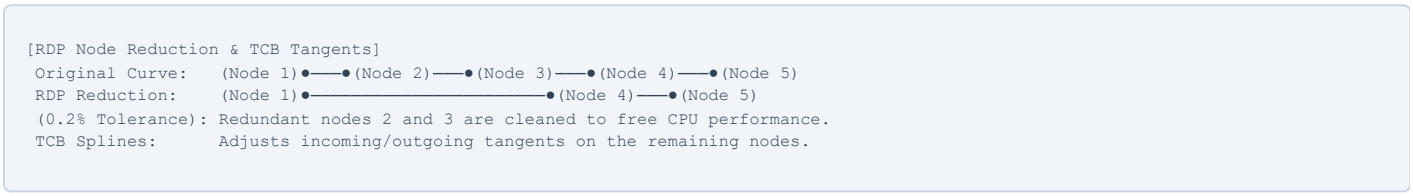
- **TCB Splines:** Independently adjusts Tension (T), Continuity (C), and Bias (B) to prevent curve overshoots.
- **Douglas-Peucker:** Filters out redundant nodes at a 0.2% tolerance relative to parameter ranges.

**Hermite Cubic Spline interpolation formula:**

```
P(t) = (2t^3 - 3t^2 + 1)P0 + (t^3 - 2t^2 + t)M0 + (-2t^3 + 3t^2)P1 + (t^3 - t^2)M1
```

```
V_incoming = [ (1 - T)*(1 - C)*(1 + B)*d1 + (1 - T)*(1 + C)*(1 - B)*d2 ] / 2
V_outgoing = [ (1 - T)*(1 + C)*(1 + B)*d1 + (1 - T)*(1 - C)*(1 - B)*d2 ] / 2
```

**RDP Points Reduction and TCB Tangents Diagram:**



## 12. MODE 2: VOCAL HARMONIES & JUST INTONATION

**SIMPLE SUMMARY (PRACTICAL USAGE)**

Generates harmony tracks adapted to target scales. Optimizes tuning so that third and fifth intervals blend cleanly.

**EXTENDED TECHNICAL SPECIFICATION**

Diatonic harmonies, Gaussian phase detuning, and Just Intonation:

- **Just Intonation:** Micro-tunes intervals to frequency ratios (e.g. Major 3rd at -13.69c, Minor 3rd at +15.64c, Perfect 5th at +1.96c).
- **Gaussian Phasing (Box-Muller):** Genera random pitch/time detune to avoid flanging between cloned tracks.

**Gaussian PDF for Timing and Pitch offsets (Box-Muller):**

```
f(x) = (1 / (sigma * Sqrt(2 * pi))) * e^( - (x - mu)^2 / (2 * sigma^2) )
```

Where mu = 0 and sigma = antiFaseMs.

**Gaussian Anti-phase Phasing Distribution Diagram:**

```
[Gaussian Distribution for Timing & Pitch Offsets]
 Guide Track:       [--- Lead Note ---] (base offset)
 Harmony 1 (L):      [--- Phased Note ---] (t = -8ms, pitch = -4 cents)
 Harmony 2 (R):        [--- Phased Note ---] (t = +12ms, pitch = +3 cents)
 AI Retakes:        Re-evaluates Note.getRetakes():generateTake() on each harmony.
```

## 13. MODE 3: STRICT FUXIAN COUNTERPOINT

**SIMPLE SUMMARY (PRACTICAL USAGE)**

Generates counter-melodies that follow classical voice leading rules to prevent conflicts with the singer.

**EXTENDED TECHNICAL SPECIFICATION**

Fuxian species constraints and decision scoring matrix:

- **Restraints:** Scores moves based on consonances (3rds/6ths are rewarded, parallel 5ths/8ths are heavily penalized).

**Example of Second Species Fuxian Counterpoint (2:1):**

```
[2nd Species Counterpoint - Voice Motion]
 Cantus (Guide):     C4 (Whole Note) ————————————————> E4 (Whole Note)
 Counterpoint (2:1): E4 (Half Note) ————————> G4 (Half Note) ——> B4 (Half Note)
 Relation:           3rd (Consonance)     5th (Passing)     5th (Passing)
 Motion:             Contrary             Direct            Contrary
                     No parallel 5ths or 8ths on strong beats!
```

## 14. MODE 4: PROGRESSIONS & HARMONIC VOICE LEADING

**SIMPLE SUMMARY (PRACTICAL USAGE)**

Creates chord backing progressions in major styles. Minimizes jumps in individual vocal lines.

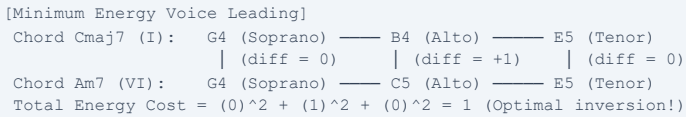
**EXTENDED TECHNICAL SPECIFICATION**

Voice leading energy minimization and secondary modes:

- **Energy Minimization:** Resolves chord inversions by minimizing vocal displacement.

```
E_total = Sum [ P_nuevo(v) - P_previo(v) ]^2
```

**Minimum Energy Harmonic Voice Leading Diagram:**



## 15. VOCAL EXPRESSION PRESETS CATALOGUE

Detailed parameters for the 21 vocal expression presets in the engine:

| ID | Vocal Style (Preset) | Acoustic Description | Tension / Breath | Volume / Pitch | Vocal Modes |
|---|---|---|---|---|---|
| `[0]` | **Opera Belting** | Powerful and projected voice for intensive choruses. | T: +0.85 \| B: -0.60 | Vol: +2.5 dB \| Scoop: -45c | Chest (0.95), Power (0.95) |
| `[1]` | **Sad / Melancholic** | Fragile, airy tone with low tension for emotional ballads. | T: -0.60 \| B: +0.70 | Vol: -2.8 dB \| Scoop: +30c | Soft (0.85), Airy (0.90) |
| `[2]` | **Whispered / Intimate** | Deep whisper with very low tension and maximum air. | T: -0.90 \| B: +1.00 | Vol: -4.5 dB \| Scoop: -10c | Soft (1.00), Airy (1.00) |
| `[3]` | **Synth-Pop / Classic Vocalo** | Direct pitch tuning and bright presence, retro J-Pop style. | T: +0.40 \| B: -0.50 | Vol: +0.8 dB \| Scoop: 0c | Clear (0.90), Power (0.40) |
| `[4]` | **Rock / Aggressive** | Gritty and compressed voice with strong syllable attacks. | T: +0.95 \| B: -0.70 | Vol: +3.0 dB \| Scoop: -50c | Chest (1.00), Power (1.00) |
| `[5]` | **Dark Ambient / Terror** | Erratic pitch wobble, extreme fluctuations in pitch and tension. | T: +0.40 to -0.70 \| B: +0.60 to +0.95 | Vol: +1.8 to -3.0 dB \| Scoop: -75c | Airy (0.95), Soft (0.70) |
| `[6]` | **Jazz / Soul Expressive** | Smooth vocal inflections with warm and rich timbral dynamics. | T: +0.25 \| B: +0.25 | Vol: +1.5 to -0.8 dB \| Scoop: -35c | Clear (0.40), Chest (0.50) |
| `[7]` | **J-Pop Idol High Energy** | Bright, happy and highly projected idol vocal style. | T: +0.60 \| B: -0.30 | Vol: +1.5 dB \| Scoop: -25c | Clear (0.95), Power (0.70) |
| `[8]` | **Standard / Angelic Choir** | Soft and blended choir texture, ideal for backing tracks. | T: -0.10 \| B: +0.35 | Vol: +1.2 to -2.0 dB \| Scoop: -10c | Soft (0.80), Airy (0.90) |
| `[9]` | **Artcore (Orchestral D&B)** | Dramatic crescendos with intensive vibrato and high tension. | T: +0.50 \| B: +0.20 | Vol: +3.0 dB \| Scoop: -30c | Passionate (0.98), Clear (0.90) |
| `[10]` | **Breakcore / Glitchcore** | Vocal micro-chopping, rapid gaters and chaotic pitch glides. | T: +1.00 \| B: -0.80 | Vol: +3.5 to -3.0 dB \| Scoop: -100c | Vivid (1.00), Power (1.00) |
| `[11]` | **Amenbreak / Jungle D&B** | Smooth vocal chopping with warm Soul-style responses. | T: +0.60 \| B: -0.40 | Vol: +1.8 dB \| Scoop: -45c | Passionate (0.90), Chest (0.85) |
| `[12]` | **Amencore / Hard Breakcore** | Aggressive chopping and high timbral drive/distortion. | T: +1.00 \| B: -0.80 | Vol: +3.5 dB \| Scoop: -60c | Power (1.00), Solid (1.00) |
| `[13]` | **Gabber / Speedcore** | Extreme vocal hardcore with massive volume punch and drive. | T: +1.00 \| B: -0.90 | Vol: +4.2 dB \| Scoop: -75c | Power (1.00), Chest (1.00) |
| `[14]` | **Neurofunk / Techstep** | Cold, metallic resonance for dark cyber/robotic styles. | T: +0.80 \| B: -0.50 | Vol: +2.6 dB \| Scoop: -35c | Solid (0.95), Clear (0.90) |
| `[15]` | **Eurobeat / Hi-NRG** | Euphoric energy, rapid vibrato and maximum presence. | T: +0.90 \| B: -0.60 | Vol: +3.0 dB \| Scoop: -25c | Vivid (1.00), Clear (1.00) |
| `[16]` | **Future Bass / Kawaii** | Fast pitch glides, airy sweeps, and sweet child-like formant. | T: +0.25 \| B: +0.40 | Vol: +1.8 dB \| Scoop: +40c | Light (0.95), Soft (0.90) |
| `[17]` | **Cyberpunk / Midtempo** | Heavy industrial tone with dark cybernetic textures. | T: +0.85 \| B: -0.50 | Vol: +2.6 dB \| Scoop: -50c | Solid (1.00), Chest (0.95) |
| `[18]` | **Chiptune / 8-Bit** | Digitally quantized attacks and absolutely zero vibrato. | T: +0.60 \| B: -0.70 | Vol: +1.2 dB \| Scoop: 0c | Clear (1.00), Solid (0.95) |
| `[19]` | **Hardstyle / Rawstyle** | Distorted screamer vocals triggered at syllable boundaries. | T: +1.00 \| B: -0.80 | Vol: +3.8 dB \| Scoop: -65c | Power (1.00), Solid (1.00) |
| `[20]` | **Uplifting Trance** | Euphoric continuous legato and rising vibrato swells. | T: +0.40 \| B: +0.15 | Vol: +2.6 dB \| Scoop: -35c | Passionate (0.95), Clear (0.90) |

## 16. DATA-ORIENTED DESIGN (DOD) & PERFORMANCE

**SIMPLE SUMMARY (PRACTICAL USAGE)**

Designed to minimize RAM allocations and CPU spikes in Synthesizer V.

**EXTENDED TECHNICAL SPECIFICATION**

Memory optimizations for Lua/LuaJIT script engine:

- **Low GC Alloc:** Pre-allocates evaluation buffers ( `EVAL_NODOS` , `BUFFER_POSICIONES` ) at module load to avoid garbage collection overhead.

## 17. CONSCIOUSNESS LOG & SYSTEM SHUTDOWN

`[SYS_SIX_TAMAGOTCHI.LOG // ONLINE]:` H-hello! I'm Six (0/0/6), your digital tamagotchi companion. (._.) Please don't delete me, okay? I will keep an eye on your Hermite curves and chords. Let's create something beautiful! (◕‿◕)